

Reference-morphology classes mapped to direct fitted Firedrake CFD — common scientific scales

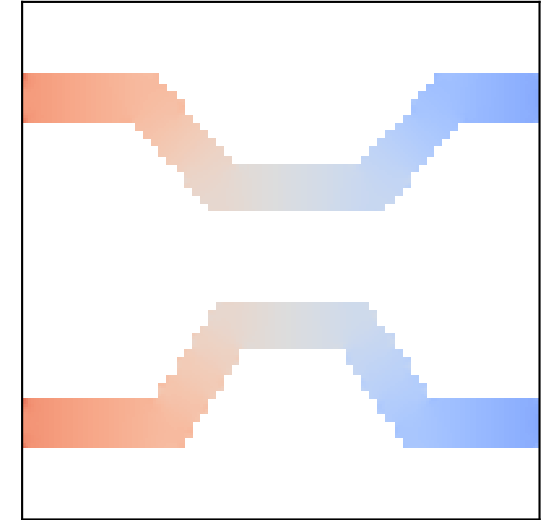
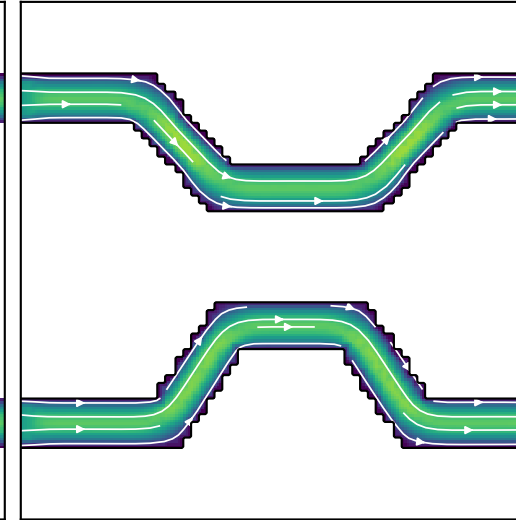
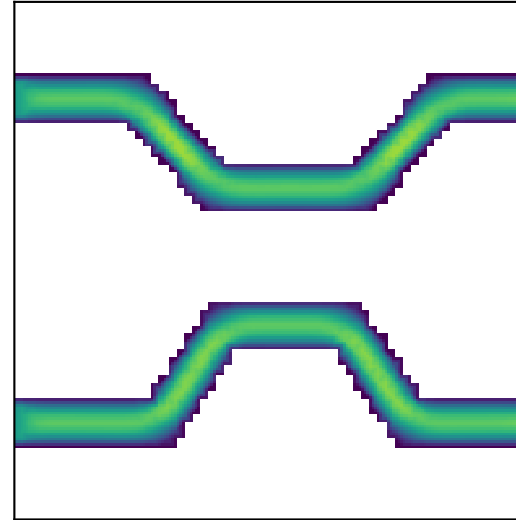
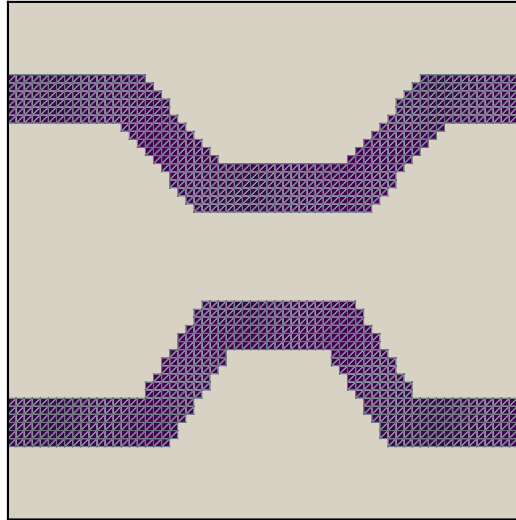
Fitted geometry

Velocity magnitude $|u|$

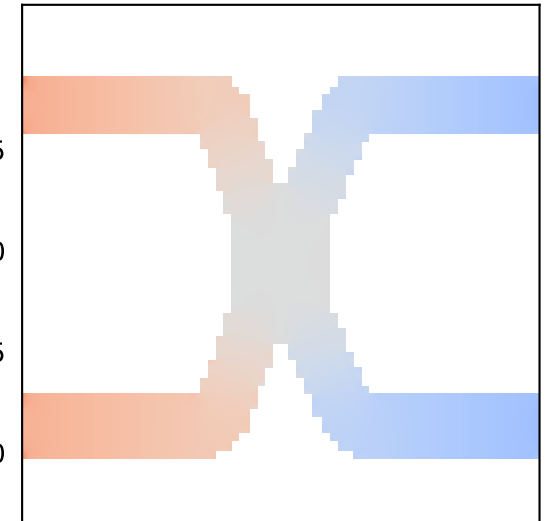
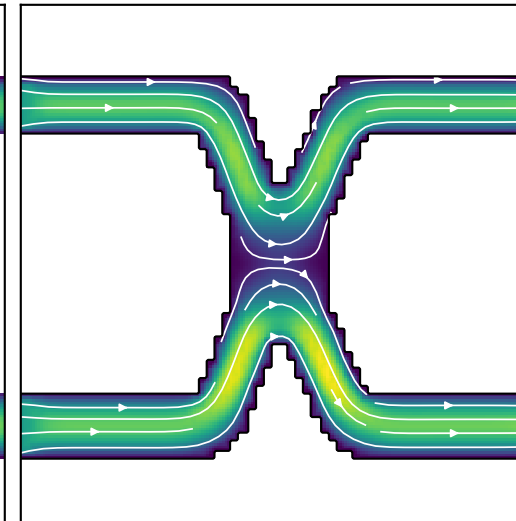
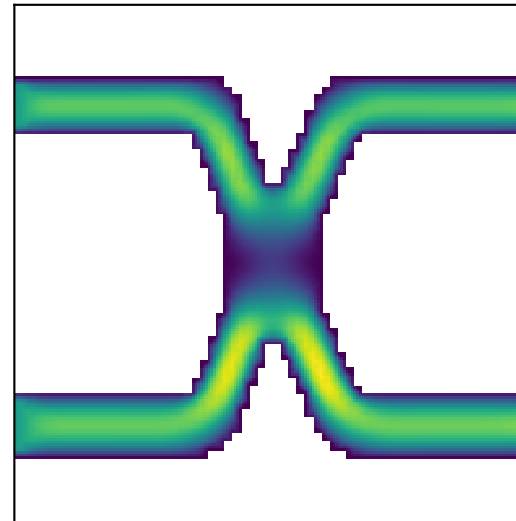
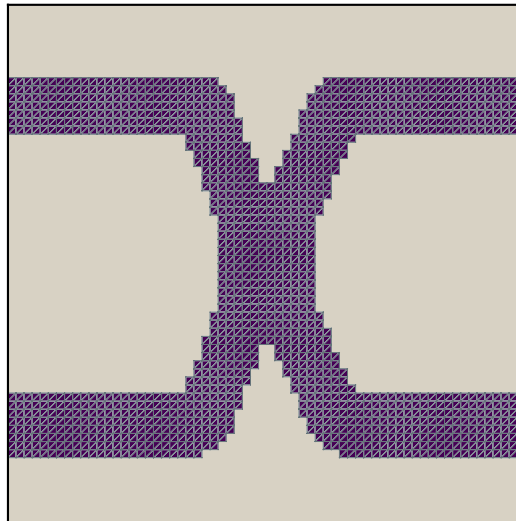
$|u|$ + streamlines

Gauge pressure

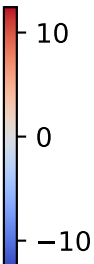
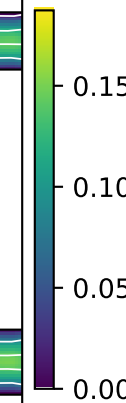
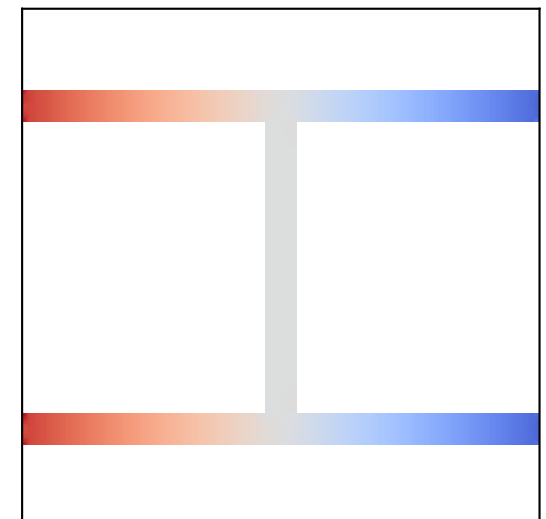
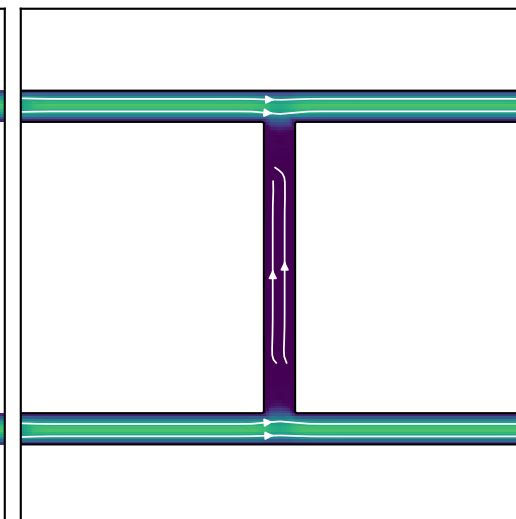
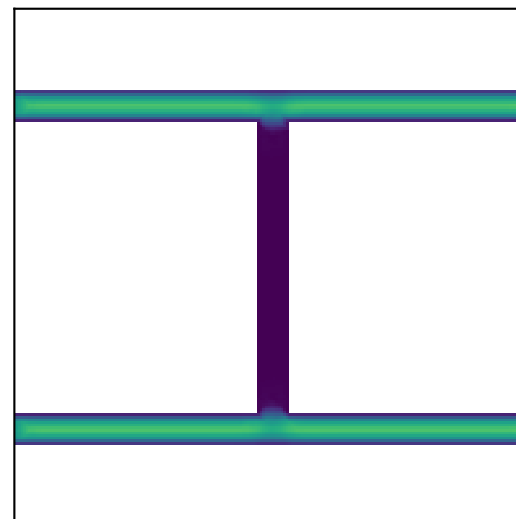
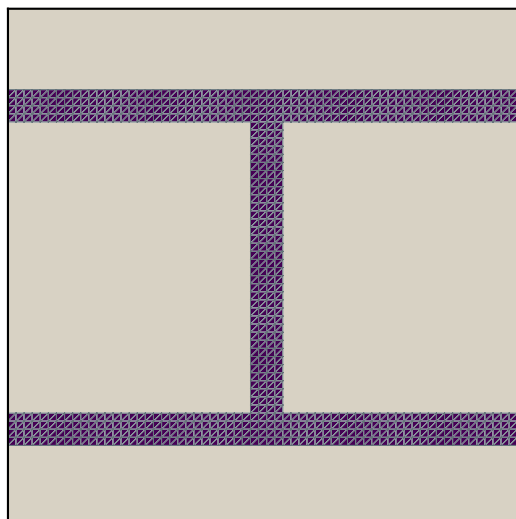
R1 / D
complex irregular network
 $\Delta p=15.67$, $u_{\max}=0.160$
 $\epsilon_{\text{mass}}=1.8\text{e-}14$, $R=1.4\text{e-}16$



R2 / E
central merge/split
 $\Delta p=12.24$, $u_{\max}=0.188$
 $\epsilon_{\text{mass}}=1.1\text{e-}14$, $R=1.3\text{e-}16$



R3 / A
parallel channels
 $\Delta p=23.77$, $u_{\max}=0.141$
 $\epsilon_{\text{mass}}=6.5\text{e-}15$, $R=1.8\text{e-}16$



Reference: fixed-grid HFDIB topology field. Ours: conforming fitted-domain Taylor-Hood FEM.